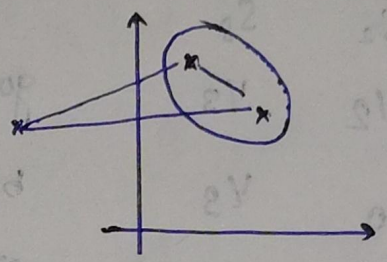
5. Word Embeddings:

In natural language processing (NLP), word embedding is a term used for the representation of words for text analysis, typically in the form of a real-valued vector that encodes the meaning of the word such that the words that are closer in the vector space are expected to be similar in meaning.

PCA

Angry Happy Excited

$\begin{bmatrix} \\ \\ \end{bmatrix}$ $\begin{bmatrix} \\ \\ \\ \end{bmatrix}$ $\leftrightarrow$ $\begin{bmatrix} \\ \\ \\ \end{bmatrix}$



word → vector

Sentence → vector

→ Vectorization

Word Embedding

Word Embedding

- Count or Frequency
  - i. OHE
  - ii. BOW
  - iii. TF-IDF
  - #Industry
- Deep learning Trained Models
  - Word2Vec
    - CBOW [Continuous Bag of words]
    - Skip grams

Frequency Models

Note: In count of frequency there are some limitations and these limitations will overcome by Deep learning Trained models.